

A Computational Method for Constructing Type E Dominating Sets of Q_{4k+1} for Larger k

Abstract

Burger and Mynhardt showed that if a dominating Type E (i, i) -set of Q_{4k+1} of cardinality $2k + 1$ exists for some fixed k , then $\gamma(Q_n) \leq \frac{3k+5}{6k+3}n + \mathcal{O}(1)$ for all sufficiently large n . Since $(3k + 5)/(6k + 3)$ decreases with k , constructing such sets for larger values of k yields stronger asymptotic upper bounds. We present a computational method for constructing Type E (i, i) -sets and use it to obtain new instances, including one with $k = 193$, the largest obtained. The method reduces the problem to finding a bijection between two prescribed multisets of diagonal labels, identifies a linear invariant satisfied by every solution, and uses this invariant both to eliminate infeasible parameter choices by an exact matching computation and to guide the search. The search uses constraint-programming initialisation, followed by simulated-annealing refinement and a targeted beam-repair procedure with diversity control based on defect patterns. The $k = 193$ construction improves the best previously known asymptotic coefficient $112/221 \approx 0.506787$ to $584/1161 \approx 0.503015$, giving $\gamma(Q_n) \leq (584/1161)n + \mathcal{O}(1)$ for all sufficiently large n .

1 Introduction

We follow the notation of [1]. The queens graph Q_n has the squares of the $n \times n$ chessboard as its vertices, two squares being adjacent if they lie in a common row, column or diagonal. A set D of squares is *dominating* if every square is in D or adjacent to a square of D , and $\gamma(Q_n)$ denotes the minimum cardinality of a dominating set.

Raghavan and Venkatesan [2] proved $\gamma(Q_n) \geq \frac{1}{2}(n - 1)$, and Weakley [3] refined this to $\gamma(Q_{4k+1}) \geq 2k + 1$. Upper bounds close to $n/2$ are harder. Burger and Mynhardt [1, Corollary 11] established the following: *if there exists a dominating (i, i) - or $(i, i + 2)$ -set of Q_{4k+1} of cardinality $2k + 1$, with $i < k$, then for all n large enough*

$$\gamma(Q_n) \leq \frac{3k + 5}{6k + 3}n + \mathcal{O}(1). \quad (1)$$

The coefficient $(3k + 5)/(6k + 3)$ is strictly decreasing in k . The bound (1) is therefore a machine that converts a single dominating set on one board Q_{4k+1} into an asymptotic bound for all large n , and the quality of the bound is governed entirely by the largest k for which such a set is known.

At the time of Burger and Mynhardt [1], the largest known Type E (i, i) -set of the required cardinality had $k = 28$, while the largest $(i, i + 2)$ -set had $k = 32$, corresponding to Q_{129} ; the latter gave $\gamma(Q_n) \leq \frac{101}{195}n + \mathcal{O}(1)$, $\frac{101}{195} \approx 0.51795$. Other classes of dominating sets and related constructions have been studied both before and since, often with substantial computational search [4, 5, 6, 7, 8]. Across these different approaches, the large structured constructions relevant to upper bounds have remained on boards of order in the low hundreds; to the best of our knowledge, none of the previously reported constructions of this kind exceeds Q_{250} . Thus, a principal obstacle to improving asymptotic upper bounds is not necessarily the absence of suitable theoretical constructions, but the difficulty of finding dominating sets satisfying their

required structure on larger boards. This is a non-trivial search problem: the number of candidate configurations grows exponentially in k , and the defining condition is a rigid exact-covering requirement admitting no slack.

This paper describes a method for this search and uses it to obtain Type E (i, i) -sets for several larger values of k , the largest being $k = 193$ on Q_{773} . By the Burger–Mynhardt bound (1), this gives the improved asymptotic upper bound $\gamma(Q_n) \leq \frac{584}{1161}n + \mathcal{O}(1)$, $\frac{584}{1161} \approx 0.503015$. Section 2 describes the notation and reduces the construction of a Type E (i, i) -set of cardinality $2k + 1$ to finding a bijection between two explicit multisets of diagonal labels, subject to a covering condition (Proposition 1). Section 3 establishes a linear invariant satisfied by every such bijection (Proposition 3) and derives an exact test that rules out infeasible parameter choices before any search is performed (Corollary 4). Section 4 describes the search method, Section 5 reconstruction and verification, and Section 6 the sets obtained. Pseudocode for every stage is collected in Appendix A.

2 Notation and reduction to a pairing problem

2.1 Board, diagonals, Type E sets

Let $n = 4k + 1$. We identify the board with a square of side n in the Cartesian plane, centred at the origin with sides parallel to the coordinate axes, and refer to squares by the coordinates of their centres. Thus, the square (x, y) lies in column x and row y . A square (x, y) is called *even* if $x + y$ is even and *odd* otherwise. We further distinguish squares according to the parities of their coordinates: a square is *even–even*, *even–odd*, *odd–even*, or *odd–odd* according as (x, y) has the corresponding coordinate parities. Rows and columns are collectively called *orthogonals*, and an orthogonal is *even* or *odd* according to the parity of its label.

We consider placements of queens on even orthogonals only, with at least one queen in each even orthogonal. Every square in an even orthogonal is then dominated, so only the odd–odd squares remain, and they must be dominated diagonally. Following [1] we pass to the *reduced board*, in which only the odd–odd squares are drawn: the square with coordinates $(2x, 2y)$ corresponds to the square (x, y) of the reduced board, whose labels run from $-k$ to k in each coordinate. All coordinates below are reduced coordinates.

Diagonals rising from left to right are *d-diagonals*, labelled by $d = y - x$; those falling are *s-diagonals*, labelled by $s = x + y$. A Type E set for which the first empty *s*-diagonals from the centre are $s = \pm i$ and the first empty *d*-diagonals are $d = \pm j$ is called an (i, j) -set. We use the following result of Burger, Cockayne and Mynhardt, quoted as Theorem 2(a) of [1]: *an (i, i) -set of queens on Q_{4k+1} lying on the diagonals*

$$s, d = 0, \pm 1, \pm 2, \dots, \pm(i - 1), \pm(i + 1), \pm(i + 3), \dots, \pm(2k - i - 1) \quad (2)$$

is dominating; conversely, an (i, i) -dominating set D with $|D| = 2k + 1$ contains queens on each of the diagonals in (2).

This paper concerns the special case of Type E (i, i) -sets.

2.2 The parameter family

Write $i = e + 1$ and $u = k - e - 1$. The positive labels occurring in (2) are

$$1, 2, \dots, e, \quad e + 2, e + 4, \dots, e + 2u,$$

which is $e + u = k - 1$ values in each of the two diagonal directions.

We seek a set of cardinality $2k + 1$ that is centrally symmetric: a queen at the centre $(0, 0)$ together with k pairs $\{(x, y), (-x, -y)\}$. Each such pair occupies the *d*-diagonals $\pm|y - x|$ and the *s*-diagonals $\pm|x + y|$, so the k pairs supply exactly k positive *d*-labels and k positive *s*-labels,

while (2) requires $k - 1$ of each. One label in each direction is therefore free. Denoting the two free labels by a and b , the multisets of positive diagonal labels used are

$$\mathbf{D} = \{1, \dots, e\} \cup \{e + 2j : j = 1, \dots, u\} \cup \{a\}, \quad \mathbf{S} = \{1, \dots, e\} \cup \{e + 2j : j = 1, \dots, u\} \cup \{b\}, \quad (3)$$

each of cardinality k . Here a and b are the free labels; they need not be new values, and may coincide with a label already present in the regular part, in which case $\mathbf{D}$ or $\mathbf{S}$ is a proper multiset. (Of the sets in Section 6 exactly one, $k = 62$, is of that kind.) A candidate configuration is determined by a bijection $\pi : \mathbf{D} \rightarrow \mathbf{S}$, which assigns to each queen pair its d -label and its s -label. The parameters of the construction are thus (k, e, a, b) , with $u = k - e - 1$ determined, and the search space is the set of bijections between $\mathbf{D}$ and $\mathbf{S}$.

2.3 Legality and the covering condition

Let (d, s) be a pair with $d \in \mathbf{D}$, $s = \pi(d) \in \mathbf{S}$, and set

$$p = \frac{d+s}{2}, \quad q = \frac{|d-s|}{2}. \quad (4)$$

Solving $y - x = \pm d$, $x + y = \pm s$ shows that the corresponding queen has coordinate magnitudes $\{|x|, |y|\} = \{p, q\}$. The pair is *legal* if

$$d \equiv s \pmod{2}, \quad d \neq s, \quad d + s \leq 2k. \quad (5)$$

The first condition makes p and q integers, the second gives $q \geq 1$, and the third gives $p \leq k$. In other words, $1 \leq q < p \leq k$, so the corresponding queen lies on the reduced board and off the centre.

A Type E set has at least one queen in each of the $2k + 1$ even orthogonals; since it has exactly $2k + 1$ queens, it has exactly one in each. In reduced coordinates this says that every row and every column contains exactly one queen. The following records when a bijection delivers that.

Proposition 1. *Let $\pi : \mathbf{D} \rightarrow \mathbf{S}$ be a bijection all of whose pairs are legal, and let (p_i, q_i) , $i = 1, \dots, k$, be given by (4). Suppose every $v \in \{1, \dots, k\}$ occurs exactly twice in the multiset $\{p_1, q_1, \dots, p_k, q_k\}$. Then the multigraph G on vertex set $\{1, \dots, k\}$ with edges $\{p_i, q_i\}$ is 2-regular, and orienting each of its cycles consistently assigns to each queen pair a coordinate pair (x, y) with $\{|x|, |y|\} = \{p_i, q_i\}$ in such a way that every nonzero row and column of the reduced board contains exactly one queen. Together with the centre, the resulting $2k + 1$ squares form a Type E (i, i) -dominating set of Q_{4k+1} with $i = e + 1$.*

The hypothesis is exactly a degree condition: each v occurring twice means each vertex of G has degree 2, so G is a disjoint union of cycles, and traversing each cycle in a fixed direction uses each v once as a row magnitude and once as a column magnitude. Domination then follows from (2), since the diagonals carried by $\mathbf{D}$ and $\mathbf{S}$ are precisely those listed there.

Accordingly, writing $c(v)$ for the number of occurrences of v in $\{p_1, q_1, \dots, p_k, q_k\}$, define

$$E(\pi) = \sum_{v=1}^k |c(v) - 2|, \quad (6)$$

so that $E(\pi) = 0$ characterises the configurations sought and $E(\pi)$ measures how far a candidate is from being one.

3 A linear invariant and an exact feasibility test

The condition $E(\pi) = 0$ is a covering condition and is not linear. It nevertheless forces a linear identity, which is the basis of both the parameter test of Section 3.1 and the steering criterion of Section 4.

Lemma 2. *For a legal pair (d, s) with p, q as in (4), $\max(d, s) = p + q$.*

Proof. For $d, s \geq 1$ we have $(d + s)/2 \geq |d - s|/2$, so $p \geq q$, and $p + q = (d + s)/2 + |d - s|/2 = \max(d, s)$. $\square$

Proposition 3. *Let π be a bijection with $E(\pi) = 0$ and put $M(\pi) = \sum_{i=1}^k \max(d_i, s_i)$. Then $M(\pi) = M^*$, where $M^* = k(k + 1)$.*

Proof. By Lemma 2, $M(\pi) = \sum_i (p_i + q_i)$. Since $E(\pi) = 0$, every $v \in \{1, \dots, k\}$ occurs exactly twice among the p_i and q_i , so this sum equals $2 \sum_{v=1}^k v = k(k + 1)$. $\square$

Proposition 3 is a necessary condition only; it says nothing about coverage. Its value is that M is a *linear* function of the bijection, whereas E is not, so M can be optimised exactly. For a candidate π we write

$$g(\pi) = |M(\pi) - M^*| \quad (7)$$

and call it the M -gap. By Proposition 3 a candidate with $g(\pi) > 0$ is not a solution, so the search treats g as a second coordinate to be driven to zero alongside E ; in practice g is closed first and then maintained (Section 4.3).

3.1 An exact infeasibility test

Corollary 4. *Fix (k, e, a, b) and let*

$$M_{\max} = \max_{\pi} \sum_i \max(d_i, s_i), \quad M_{\min} = \min_{\pi} \sum_i \max(d_i, s_i),$$

the extrema being over bijections $\pi : \mathbf{D} \rightarrow \mathbf{S}$ all of whose pairs are legal. If $M^ > M_{\max}$ or $M^* < M_{\min}$, then no bijection with $E(\pi) = 0$ exists for these parameters.*

Proof. Immediate from Proposition 3. $\square$

Both extrema are maximum- and minimum-weight perfect matchings in the bipartite graph on $\mathbf{D} \times \mathbf{S}$ whose edges are the legal pairs, with edge weights $\max(d, s)$. They are computed exactly by the Hungarian algorithm in time $\mathcal{O}(k^3)$ — under one second for the values of k considered here — so the verdict is a proof, not an estimate. See Algorithm 1.

Remark (The test is one-sided.). $M_{\min} \leq M^* \leq M_{\max}$ does *not* imply that a bijection attaining M^* exists, still less one with $E = 0$; it asserts only that the necessary condition of Proposition 3 is not violated.

3.2 An empirical observation on scale

This empirical observation is used only heuristically in Section 4.2. For each verified dominating set of the above type, normalise the diagonal labels of its queens by k , i.e. consider the points $(d/k, s/k)$. Over the ten verified sets with k ranging from 62 to 193, the correlation of these two coordinates lies in $[-0.404, -0.376]$, with pooled value -0.390 . The macroscopic distribution of $(d/k, s/k)$ thus appears to be nearly independent of k .

4 The search method

The method maintains a current bijection β and improves it in rounds; each round applies a stochastic refinement stage followed by a beam repair stage. All comparisons between bijections use the lexicographic key

$$\text{key}(\pi) = (E(\pi), g(\pi), R_3(\pi)), \quad R_3(\pi) = \left| \sum_{v=1}^k v^3 (c(v) - 2) \right|, \quad (8)$$

with E from (6) and g from (7). Note that R_3 is the modulus of a *signed* sum: deficits and surpluses cancel, so R_3 is small when the defects are balanced across the range $1, \dots, k$ and large when they are lopsided. Its role is discussed in Section 4.4. Pseudocode for the whole method is Algorithm 5.

4.1 Parameter selection

Given a target k , we scan odd values of e downward in a neighbourhood of $0.42k$. For each e , several candidate free-label pairs (a, b) of parity opposite to e are sampled; values already present in the regular part are allowed, since $\mathbf{D}$ and $\mathbf{S}$ are multisets. Each sampled quadruple (k, e, a, b) is tested by Corollary 4, and those ruled out are discarded. The largest tested e with at least one surviving pair is retained, and the surviving pairs are briefly initialised and refined before the best is selected. See Algorithm 2.

The feasibility filter is a proof-level test for each fixed quadruple (k, e, a, b) and is inexpensive; omitting it risks committing the search to parameters for which no configuration exists. We scan e from large to small in a neighbourhood of $0.42k$, motivated by the empirical observation that the verified sets satisfy $e/k \in [0.403, 0.430]$ (Table 1). Under the same search budget, choices with $e/k \approx 0.39$ reach only $E \approx 58$, compared with $E \approx 12$ – 16 for $e/k \approx 0.42$.

4.2 Initialisation

Two initialisations were used.

(a) Constraint programming. Model the bijection with a Boolean variable x_{ds} for each legal pair, subject to $\sum_s x_{ds} = 1$ for each d and $\sum_d x_{ds} = 1$ for each s ; introduce $c(v)$ and $|c(v) - 2|$ as auxiliary integer variables; and *minimise* E . We found that this minimisation formulation produces useful approximate solutions, reducing E from $\mathcal{O}(k)$ into the tens (for $k = 250$, $E = 36$ in 200 s).

(b) Assignment against the limit shape (optional). Estimate the density $\hat{P}$ of $(d/k, s/k)$ from known sets (Section 3.2), give each legal pair the cost $-\log \hat{P}(d/k, s/k)$, and compute a minimum-cost perfect matching by the Hungarian algorithm. To place the result on the shell of Proposition 3, add a Lagrangian term $\lambda \max(d, s)$ to the cost and bisect on λ ; M is non-increasing in λ , so the bisection is well defined. See Algorithm 3.

The new sets of Section 6 were obtained with initialisation (a); initialisation (b) was developed subsequently and validated separately, as follows. Initialisation (b) costs about one second and yields a bijection with large E (≈ 350 – 400) but small M -gap and the correct macroscopic shape. Under identical subsequent budgets it performs at least as well as (a): on $k = 250$ with two seeds, (b) gave final $E = 14, 14$ and (a) gave $E = 16, 14$. Note that the initial value of E is a poor predictor of the final value, while the macroscopic structure appears to be more informative.

4.3 Local refinement

Two independent simulated-annealing runs, with different seeds and perturbation strengths, are started from the current bijection. Their role is to reach the shell $M = M^*$ and reduce E ; in practice, the M -gap is brought to within one unit within seconds, and E into the range 10–16. Note that a run terminating with g in the hundreds or thousands typically indicates misspecified parameters rather than slow progress.

4.4 Beam repair

The refined bijections, *together with the incumbent best bijection* β , seed a beam search (Algorithm 4) that maintains a frontier of several thousand distinct candidates and expands each by three families of moves.

Targeted transpositions. For an under-covered value v and a position i , the only labels s' for which position i would cover v are

$$s' \in \{2v - d_i, d_i + 2v, d_i - 2v\},$$

by (4): the first makes $p_i = v$, the others $q_i = v$. For each such s' held by some position j , the transposition of the two labels is generated whenever both resulting pairs remain legal. This family is enumerated exhaustively over under-covered values and positions.

Targeted 3-cycles. Pairs (i, j) produced above are extended by a third position l , drawn with probability 0.8 from positions whose p - or q -value is over-covered and otherwise uniformly from all positions; both cyclic orientations are generated whenever the resulting pairs are legal. For each candidate we make $\min\{70, \max\{25, \lfloor |\mathcal{T}|/4 \rfloor\}\}$ such attempts, where $\mathcal{T}$ is the set of targeted position pairs.

Uniform transpositions. In addition, eight uniformly sampled legal transpositions are generated per candidate to provide background diversity.

Candidates are deduplicated, those whose E exceeds the incumbent by more than a slack τ are discarded, and the survivors are ordered by key as in (8). The frontier is then filled by admitting at most κ candidates per *defect signature*

$$\text{sig}(\pi) = \{(v, c(v)) : c(v) \neq 2\},$$

before any remaining slots are filled in key order. This quota is what keeps the frontier genuinely diverse rather than occupied by minor variants of a single defect pattern. We used $\tau = 10$ and $\kappa = 5$.

Because g is the second component of key, the M -gap is taken into account at every beam depth; in particular, among candidates with the same value of E , those closer to the shell $M = M^*$ are preferred. The final component, R_3 , is used mainly as an empirical progress indicator. In our experiments, continued variation of R_3 across depths was associated with an active search basin, whereas a value that remained unchanged for roughly 20 consecutive depths typically indicated stagnation. In the successful $k = 193$ run, R_3 fell $180 \rightarrow 120 \rightarrow 0$ in the depths immediately preceding $E = 0$. Accordingly Algorithm 4 returns, besides the best bijection, a Boolean flag LIVE indicating whether the frontier leader's R_3 changed over the final 20 depths.

Beam memory grows with width and with k : at $k = 250$ a width of 4000 exhausted 6 GB and was terminated by the operating system, while width 1800 required 2.5 GB.

4.5 Round control and basin concentration

A round is accepted only if it does not increase key; otherwise it is discarded and the next round resumes from the retained state. Accepting on E alone is insufficient: a round tying on E while

degrading the M -gap moves the search off the shell, and was observed to replace a state with $g = 1$ by one with $g = 39$.

Rounds alone reach $E \in \{6, \dots, 12\}$ on the shell but do not reliably terminate. The following step was required to complete the $k = 193$ construction and is the operative part of the method at low E .

Concentration step. Select a state that (i) lies on the shell, (ii) has $E \leq 8$, and (iii) has R_3 *still varying*. Retain it as the reference state. Generate several refinement runs *initialised at that state*, with distinct seeds and perturbation strengths, rather than restarting from the round frontier. Then execute one beam pass of increased width and depth over the union of the reference state and those runs.

We use criterion (iii) as an empirical indicator of whether the basin remains active; in our experiments, states with equal E but constant R_3 were generally unresponsive to further search. The concentration step is included in Algorithm 5.

5 Reconstruction and verification

A bijection with $E = 0$ is converted to a configuration by forming the multigraph G of Proposition 1, orienting its cycles, recovering signed coordinates from the magnitudes, rescaling to the unreduced board, and adjoining the centrally symmetric images and the centre, giving $2k + 1$ squares (Algorithm 6).

Each configuration is verified three times independently: (i) by direct enumeration over all n^2 squares, checking that each is occupied or attacked; (ii) by an independently implemented checker applied to the coordinate list; and (iii) by a checker that reads only the coordinates, re-derives the structural parameters from the geometry, confirms the set is Type E of the required form, and computes the resulting coefficients. Test (iii) is the substantive one, since it relies on no quantity recorded by the search.

6 Results

Table 1 lists ten configurations of this family found by our method and verified independently. Every one satisfies $E = 0$ and, as Proposition 3 requires, $M - M^* = 0$ exactly. The resulting bounds are reported in the final column.

n	k	e	u	a	b	$(3k + 5)/(6k + 3)$
249	62	25	36	19*	79*	$191/375 \approx 0.509333$
285	71	29	41	74	74	$218/429 \approx 0.508159$
401	100	43	56	164	186	$305/603 \approx 0.505804$
421	105	45	59	170	190	$320/633 \approx 0.505529$
509	127	53	73	116	184	$386/765 \approx 0.504575$
521	130	55	74	146	238	$395/783 \approx 0.504470$
561	140	59	80	176	222	$425/843 \approx 0.504152$
569	142	59	82	120	170	$431/855 \approx 0.504094$
593	148	63	84	200	284	$449/891 \approx 0.503928$
773	193	81	111	236	250	$584/1161 \approx 0.503015$

Table 1: Verified Type E (i, i) -sets found by our method, with $i = e + 1$; in every case the dominating set has cardinality $2k + 1$ and $M - M^* = 0$. *: for $k = 62$ the free labels coincide with labels already present in the regular part, so **D** and **S** are proper multisets (Section 2.2).

Every configuration in Table 1 was independently verified using the three checks described in Section 5. For the largest instance, $n = 773$, substituting $k = 193$ into (1) gives

$$\gamma(Q_n) \leq \frac{584}{1161} n + \mathcal{O}(1), \quad \frac{584}{1161} \approx 0.503015,$$

for all sufficiently large n . This improves the recent asymptotic coefficient $112/221 \approx 0.506787$ obtained by Kong [8].

Computational effort. Computations were performed on an Intel Xeon 2.80 GHz machine with two available processors. The two simulated-annealing searches were run concurrently, whereas beam repair was single-threaded. For the smallest reported instance, $n = 249$, the construction was obtained in about 31 seconds. The $n = 561$ and $n = 593$ constructions took about 12 and 10 minutes, respectively, while the largest instance, $n = 773$, took about 29 minutes. The $n = 249$ instance was completed by simulated annealing directly from the constraint-programming initialization, while $n = 593$ was completed automatically after two SA-beam cycles. The $n = 561$ and $n = 773$ instances required additional continuation and repair, with the latter ultimately completed by the concentration step.

Certificates and verification. The certificates for the constructions reported in Table 1, together with a Python verifier, are publicly available at <https://github.com/TahaRostami/Gamma/tree/upper-bound>.

7 Conclusion

We have presented a computational method for constructing Type E (i, i) -dominating sets of Q_{4k+1} with cardinality $2k + 1$. The method reduces the construction problem to a constrained pairing of diagonal labels, identifies a linear invariant satisfied by every solution, and uses this invariant both for an exact matching-based infeasibility test and for guiding the subsequent search. Constraint-programming initialisation, stochastic refinement, and targeted beam repair are then used to construct the required bijections.

Using this approach, we obtained and independently verified Type E constructions for several values of k , the largest being $k = 193$ on Q_{773} . By the construction of Burger and Mynhardt, this yields $\gamma(Q_n) \leq \frac{584}{1161} n + \mathcal{O}(1) \approx 0.503015 n + \mathcal{O}(1)$ for all sufficiently large n , improving the recent asymptotic coefficient $112/221 \approx 0.506787$ obtained by Kong.

The search remains heuristic and stochastic, particularly in completing low-error states. We also investigated whether the computed configurations exhibit smaller structural motifs that could be used to construct larger solutions directly, but did not identify such a structure. Improving the low-error repair stage and searching automatically for reusable structural patterns are natural directions for further work. Since $(3k + 5)/(6k + 3)$ approaches $1/2$ as k grows, still larger Type E constructions would yield asymptotic upper bounds progressively closer to $1/2$.

References

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A Algorithms

Throughout, a bijection $\pi : \mathbf{D} \rightarrow \mathbf{S}$ is stored as an array $s[1..k]$ with $s_i = \pi(d_i)$; p_i, q_i are as in (4); $c(v)$ is the number of occurrences of v among $p_1, q_1, \dots, p_k, q_k$; E, g, R_3 and key are as in (6), (7) and (8); and $\text{sig}(\pi)$ is the defect signature of Section 4.4. $\text{INITIALIZE}(k, e, a, b)$ denotes either initialisation described in Section 4.2, and $\text{ANNEAL}(\pi, \text{seed}, \text{kick})$ denotes a simulated-annealing run started at π (Section 4.3).

Algorithm 1 $\text{ADMISSIBLE}(k, e, a, b)$ — exact parameter test (Corollary 4)

Input: k, e, a, b , with $1 \leq e \leq k - 2$

Output: INFEASIBLE (a proof) or ADMISSIBLE (necessary condition only)

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1:  $u \leftarrow k - e - 1$ 
2: build  $\mathbf{D}, \mathbf{S}$  as in (3)
3:  $L \leftarrow \{(d, s) \in \mathbf{D} \times \mathbf{S} : (d, s) \text{ legal by (5)}\}$ 
4:  $w(d, s) \leftarrow \max(d, s)$ 
5:  $M_{\max} \leftarrow \text{MAXWEIGHTPERFECTMATCHING}(\mathbf{D}, \mathbf{S}, L, w)$   $\triangleright$  Hungarian,  $\mathcal{O}(k^3)$ , exact
6:  $M_{\min} \leftarrow \text{MINWEIGHTPERFECTMATCHING}(\mathbf{D}, \mathbf{S}, L, w)$ 
7: if  $M^* > M_{\max}$  or  $M^* < M_{\min}$ , where  $M^* = k(k + 1)$  then
8:   return INFEASIBLE
9: end if
10: return ADMISSIBLE

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Algorithm 2 SELECTPARAMETERS($k; m$)

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1: for  $e \leftarrow$  largest odd integer  $\leq 0.44k$  downto  $\lceil 0.38k \rceil$  do
2:    $\Gamma \leftarrow$  positive labels of parity opposite to  $e$ 
3:    $\mathcal{A} \leftarrow \emptyset$ 
4:   for  $t \leftarrow 1$  to  $m$  do
5:     sample  $\{a_t, b_t\} \subset \Gamma$ 
6:      $verdict \leftarrow$  ADMISSIBLE( $k, e, a_t, b_t$ )
7:     if  $verdict$  is ADMISSIBLE then
8:        $\pi_t^0 \leftarrow$  INITIALIZE( $k, e, a_t, b_t$ )
9:        $\pi_t \leftarrow$  ANNEAL( $\pi_t^0$ , short budget)
10:       $\mathcal{A} \leftarrow \mathcal{A} \cup \{(t, \pi_t)\}$ 
11:    end if
12:  end for
13:  if  $\mathcal{A} \neq \emptyset$  then
14:     $t^* \leftarrow \arg \min_{(t, \pi_t) \in \mathcal{A}} \text{key}(\pi_t)$ 
15:    return  $(e, a_{t^*}, b_{t^*})$   $\triangleright$  largest tested  $e$  with a surviving pair
16:  end if
17: end for
18: return FAILURE  $\triangleright$  no sampled quadruple survived in the search range
```

Algorithm 3 LIMITSHAPEINIT($k, e, a, b; \hat{P}, T, \Lambda$) — on-shell initialisation

```
1:  $C(d, s) \leftarrow -\log \hat{P}(d/k, s/k)$  for  $(d, s) \in L$ , and  $+\infty$  otherwise  $\triangleright \hat{P}$  from Section 3.2
2:  $w(d, s) \leftarrow \max(d, s)$ ;  $\lambda_{\text{lo}} \leftarrow -\Lambda$ ;  $\lambda_{\text{hi}} \leftarrow +\Lambda$ ;  $\pi^* \leftarrow \perp$ ;  $g^* \leftarrow +\infty$ 
3: for  $t \leftarrow 1$  to  $T$  do
4:    $\lambda \leftarrow (\lambda_{\text{lo}} + \lambda_{\text{hi}})/2$ 
5:    $\pi \leftarrow$  HUNGARIAN( $C + \lambda w$ )
6:    $M \leftarrow \sum_i w(d_i, \pi(d_i))$ 
7:   if  $|M - M^*| < g^*$  then
8:      $\pi^* \leftarrow \pi$ ;  $g^* \leftarrow |M - M^*|$ 
9:   end if
10:  if  $M > M^*$  then  $\triangleright M$  is non-increasing in  $\lambda$ , so raise the penalty
11:     $\lambda_{\text{lo}} \leftarrow \lambda$ 
12:  else
13:     $\lambda_{\text{hi}} \leftarrow \lambda$ 
14:  end if
15: end for
16: return  $\pi^*$   $\triangleright g(\pi^*)$  small by construction;  $E(\pi^*)$  large
```

Algorithm 4 BEAMREPAIR($\mathcal{W}; W, \Delta, \tau, \kappa$) — diverse targeted beam

Input: warm set $\mathcal{W}$, width W , depth Δ , slack τ , quota κ **Output:** best bijection β , and a flag LIVE

```
1:  $B \leftarrow \mathcal{W}$ ;  $\beta \leftarrow \arg \min_{\pi \in B} \text{key}(\pi)$ 
2: for  $\delta \leftarrow 1$  to  $\Delta$  while  $E(\beta) > 0$  do
3:    $\mathcal{C} \leftarrow \emptyset$ 
4:   for all  $\pi \in B$  do
5:      $U \leftarrow \{v : c(v) < 2\}$ ;  $X \leftarrow \{i : p_i \text{ or } q_i \text{ is over-covered under } \pi\}$ 
6:      $\mathcal{T} \leftarrow \emptyset$ 
7:     for all  $v \in U$ , positions  $i$ , and  $s' \in \{2v - d_i, d_i + 2v, d_i - 2v\}$  do
8:       for all  $j$  with  $\pi(d_j) = s'$  do
9:         if transposing  $s_i$  and  $s_j$  keeps both resulting pairs legal then
10:           $\mathcal{T} \leftarrow \mathcal{T} \cup \{(i, j)\}$ ;  $\mathcal{C} \leftarrow \mathcal{C} \cup \{\pi \cdot (i j)\}$ 
11:        end if
12:      end for
13:    end for
14:    for  $t \leftarrow 1$  to  $\min(70, \max(25, \lfloor |\mathcal{T}|/4 \rfloor))$  do ▷ targeted 3-cycles
15:      draw  $(i, j)$  uniformly from  $\mathcal{T}$ 
16:      draw  $l$  from  $X$  with probability 0.8, and otherwise uniformly from all positions
17:      add to  $\mathcal{C}$  both cyclic perms. of  $(s_i, s_j, s_l)$  for which all resulting pairs are legal
18:    end for
19:    add to  $\mathcal{C}$  eight uniformly sampled legal transpositions of  $\pi$  ▷ background diversity
20:  end for
21:  delete from  $\mathcal{C}$  duplicate bijections and every  $\pi$  with  $E(\pi) > E(\beta) + \tau$ 
22:  sort  $\mathcal{C}$  increasingly by key
23:   $B \leftarrow$  the first  $W$  candidates in this order, initially admitting at most  $\kappa$  candidates per
  defect signature sig
24:  if fewer than  $W$  candidates were admitted, fill the remaining places from  $\mathcal{C}$  in key order
25:   $\beta \leftarrow \arg \min_{\pi \in B \cup \{\beta\}} \text{key}(\pi)$ 
26:  record  $R_3$  of the current frontier leader
27: end for
28: LIVE  $\leftarrow$  true if the recorded  $R_3$  value changed during the final 20 depths, and false
  otherwise
29: return  $(\beta, \text{LIVE})$ 
```

Algorithm 5 SEARCH(k) — main search loop

Input: board parameter k **Output:** a Type E configuration, or FAILURE

```
1:  $(e, a, b) \leftarrow \text{SELECTPARAMETERS}(k; m)$ 
2: if  $(e, a, b) = \text{FAILURE}$  then
3:   return FAILURE
4: end if
5:  $\pi^0 \leftarrow \text{INITIALIZE}(k, e, a, b)$ 
6:  $\beta \leftarrow \text{ANNEAL}(\pi^0)$ 
7: for  $r \leftarrow 1$  to  $R$  do
8:    $\pi_1 \leftarrow \text{ANNEAL}(\beta, \text{seed}_{r,1}, \text{kick}_{r,1})$ 
9:    $\pi_2 \leftarrow \text{ANNEAL}(\beta, \text{seed}_{r,2}, \text{kick}_{r,2})$  ▷ independent runs
10:   $(\pi, \text{LIVE}) \leftarrow \text{BEAMREPAIR}(\{\beta, \pi_1, \pi_2\}; W, \Delta, \tau, \kappa)$ 
11:  if  $\text{key}(\pi) \leq \text{key}(\beta)$  then
12:     $\beta \leftarrow \pi$ 
13:  end if
14:  if  $E(\beta) = 0$  then
15:    return RECONSTRUCT( $\beta$ )
16:  end if
17:  if  $E(\beta) \leq 8$  and  $g(\beta) = 0$  and LIVE then ▷ concentration step
18:     $\mathcal{A} \leftarrow \emptyset$ 
19:    for  $j \leftarrow 1$  to  $m'$  do
20:       $\pi'_j \leftarrow \text{ANNEAL}(\beta, \text{seed}'_j, \text{kick}'_j)$ 
21:       $\mathcal{A} \leftarrow \mathcal{A} \cup \{\pi'_j\}$ 
22:    end for
23:     $(\pi, \cdot) \leftarrow \text{BEAMREPAIR}(\{\beta\} \cup \mathcal{A}; W', \Delta', \tau, \kappa)$ 
24:    if  $\text{key}(\pi) \leq \text{key}(\beta)$  then
25:       $\beta \leftarrow \pi$ 
26:    end if
27:    if  $E(\beta) = 0$  then
28:      return RECONSTRUCT( $\beta$ )
29:    end if
30:  end if
31: end for
32: return FAILURE
```

Algorithm 6 RECONSTRUCT(π) and verification

Input: a bijection π with $E(\pi) = 0$

```
1:  $G \leftarrow$  the multigraph on  $\{1, \dots, k\}$  with edges  $\{p_i, q_i\}$  ▷ 2-regular by Proposition 1
2: for all cycles  $C$  of  $G$  do
3:   traverse  $C$  in a fixed direction, assigning the roles  $(|x|, |y|) = (p_i, q_i)$  consistently along  $C$ 
4: end for
5: recover signed  $(x, y)$  by the sign choice satisfying  $|x - y| = d_i$  and  $|x + y| = s_i$ 
6: rescale to the unreduced board; adjoin  $(-x, -y)$  for every queen, and the centre
7: verify: (i) sweep all  $n^2$  squares; (ii) an independently implemented checker; (iii) a checker re-deriving the structural parameters from the geometry alone
8: return the  $2k + 1$  squares
```
